EEE102 – Elements of Electrical Engineering

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Unit- II: Analysis of AC Circuits

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1.1 Introduction

The alternating system in which the magnitudes of the voltage and of the current vary in a repetitive manner. Examples of such repetitive currents are shown in Fig. 1.21. It flows first in one direction and then in the other. The cycle of variation is repeated exactly for each direction.

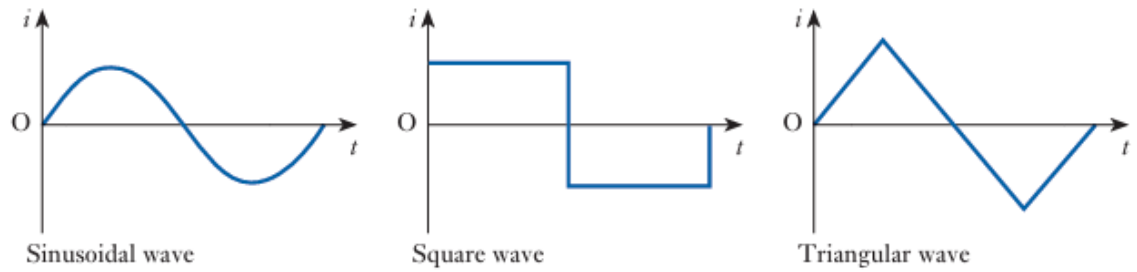


Fig. 1.21. Alternating current waveforms.

Instantaneous value: The magnitude of a waveform at any instant in time with reference to zero. Instantaneous values are denoted by lower case symbols such as e, v & i etc.

1.2. Concept of Average and RMS values

Average Value (Iav)

This is defined on the basis of amount of charge transferred, which is given by q = It. The average value of an alternating current is equal to that (D.C) steady current, which transfers the same amount of charge as transferred by the alternating current across the same circuit and in the same time.

Or, alternatively, average value of current is

$$\frac{\text{Area enclosed over half-cycle}}{\text{Length of base over half-cycle}} \qquad (1)$$

In the case of sinusoidal wave, the average value is obtained by adding or integrating the instantaneous values of the quantity over one alternating or 1/2 cycles only. Consider a sinusoidal varying alternating current $\mathrm{i}= I_m \sin\theta$ as shown in Fig. 1.22.

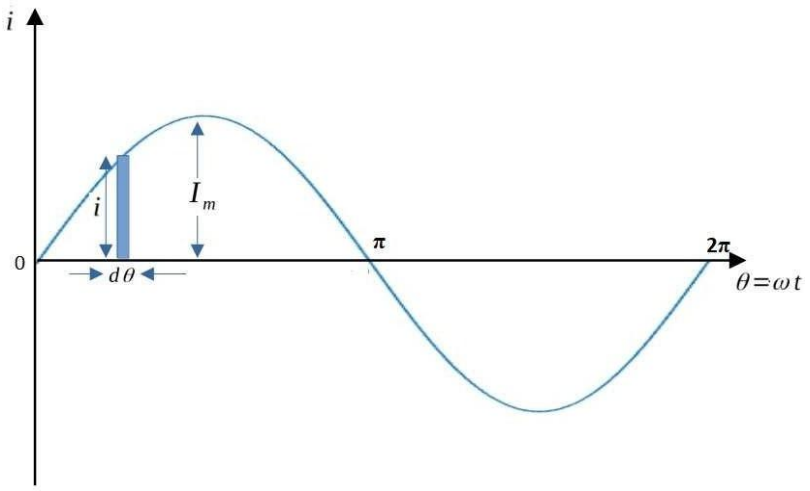


Figure 1.22: Average Value

For a very small interval dθ radians, the area of the shaded strip is i dθ ampere radians. The use of the unit 'ampere radian 'avoids converting the scale on the horizontal axis from radians to seconds, therefore, total area enclosed by the current wave over half-cycle is

$$\int_0^{\pi} id\theta = \int_0^{\pi} I_m sin\theta = I_m[-cos\theta]_0^{\pi} = 2I_m \text{ ampere radians}$$

average value of current over a half cycle is $\frac{2I_m}{\pi} = 0.637I_m$

Thus the average value of sinusoidal quantity is 0.637 times its maximum value.

Note: The average value over full cycle of sinusoidal wave is zero.

Root Mean Square (RMS) value or Effective value: (I):

In a.c., the average value is of comparatively little importance. This is due to the fact that it is the power produced by the electric current that usually matters. Thus, if a sinusoidal varying alternating current i= I_m sinθ is passed through a resistor having resistance R ohms, the heating effect of i is i^2R.

Thus the RMS value of an alternating current is defined as that steady(D.C) current which when flowing through a given resistor for a given time produces the same amount of heat as produced by the alternating current when flowing through the same resistor for the same period of time.

The average heating effect can be expressed as follows:

Average heating effect over half-cycle is

$$\frac{\text{Area enclosed by } i^2R \text{ over half-cycle}}{\text{Length of base}} \tag{1.10}$$

This is a more convenient expression to use when deriving the r.m.s. value of a sinusoidal current. If the current is passed through a resistor having resistance R ohms, instantaneous heating effect = i^2 R watts. The variation of i^2R during a complete cycle is shown in Fig 1.23 below. During interval dθ radians, heat generated

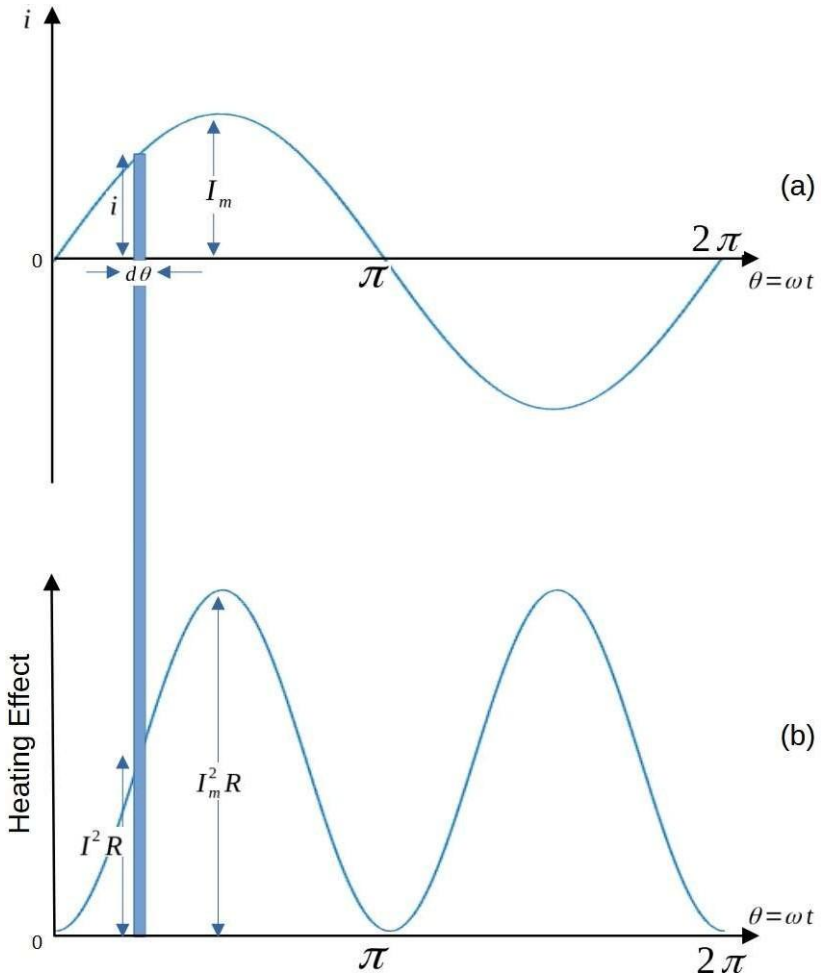


Figure 1.23: RMS Value

is i^2R dθ watt radians and is represented by the area of the shaded strip. Hence heat generated during the first half-cycle is area enclosed by the i^2R curve and is

$$\begin{aligned}\int_0^{\pi} i^2 R d\theta &= I_m^2 R \int_0^{\pi} sin^2\theta d\theta \\ &= I_m^2 R \int_0^{\pi} (\frac{1-cos2\theta}{2}) \\ &= I_m^2 R \left[\frac{\theta}{2} - \frac{sin2\theta}{4}\right]_0^{\pi} \\ &= \pi R \frac{I_m^2}{2}\end{aligned}$$

From Eq. 1.10 average heating effect is

$$= \frac{\pi R \frac{I_m^2}{2}}{\pi} = R\frac{I_m^2}{2}$$

If I is the value of direct current through the same resistance to produce the same heating effect

$$I^2 R = R\frac{I_m^2}{2}$$
$$I = \frac{I_m}{\sqrt{2}} = 0.707 I_m$$

1.2.3 Form Factor

The ratio of RMS value to average value of an alternating quantity is called form factor. For sine wave current,

$$K_f = \frac{I}{I_{av}} = \frac{I_m/\sqrt{2}}{2I_m/\pi} = 1.11$$

1.2.4 Peak Factor

The ratio of maximum value to the RMS value of an alternating quantity is called peak factor For sine wave current,

$$K_p = \frac{I_m}{I_{rms}} = \frac{I_m}{I_m/\sqrt{2}} = 1.414$$

Numericals

1. An alternating voltage has the equation v =141.4 sin377t Volts. what are the values of (a) r.m.s voltage (b) frequency (c) the instantaneous voltage when t=3ms.

   Solution:

   Given v =141.4 sin377t

   Comparing this equation with v = V_m sinω t

   ω= angular velocity =2πf radians / second

   a) r.m.s voltage = V = 0.707 × V_m

   V_m= 141.4V

   V= 0.707×141.4 = 99.96V.

   Angular velocity =ω = 2πf , since ω= 377.

   b) Frequency f=$\omega/2\pi$ =377/2π= 60Hz.

   Given t=3ms.

   c)Instantaneous voltage v= $V_m$sinωt

   v=141.4 sin (377×3×10-3× 180/π) = 127.94 V (Keep the calculator in degree mode).

2. The maximum current in a sinusoidal a.c. circuit is 10A. what is the instantaneous current at 45°

   Solution:

   i = I_m sinθ I_m=10A and θ=45°
   i=10x sin45°=7.07A

3. An alternating current of sinusoidal waveform has an r.m.s. value of 10.0 A. What are the peak values of this current over one cycle?

   Solution:

   I_m= I / 0.707 =10 / 0.707 = 14.14 A

   The peak values therefore are 14.14 A and -14.14 A.

1.3 A.C. Circuits

Now that we are familiar with alternating currents and voltages, we can apply them in turn to each of the three basic components of a circuit, i.e. to a resistor, to an inductor and to a capacitor. We shall find that each responds in a completely different manner with the result that the current and voltage do not rise and fall at the same time unless the circuit only contains resistance. For inductors and capacitors, the relationship between voltage and current is termed the reactance and we find that in practice most circuits contain both resistance and reactance. We shall therefore look at circuits in which both resistance and reactance appear in series.

1.3.1 Fundamentals of AC Circuits

Capacitance (C)

A capacitor is a device which can store electric charge for short periods of time. Like resistors, capacitors can be connected in series and parallel.
The property of a capacitor to store electric charge when its plates are at different potentials is referred to as its *capacitance.*
The unit of capacitance is farad (abbreviation F) which may be defined as the *capacitance of a capacitor between the plates of which there appears a potential difference of 1 volt when it is charged by 1 coulomb of electricity.*
The circuit symbol of capacitor is shown in Figure 1.25.

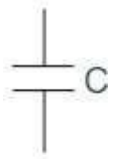


Figure 1.25: Capacitance symbol and notation

The capacitance of the capacitor is given by

$$Capacitance = \frac{\text{Charge}}{\text{Applied p.d}} = \frac{Q}{V} \tag{1.11}$$

$$Q = CV \tag{1.12}$$

In practice, the farad is found to be inconveniently large and the capacitance is usually expressed in *microfarads*(μF) or in *picofarads* (pF) where
$1\mu F = 10^{-6}F$ and $1pF = 10^{-12}F$

Inductance (L)

Inductance is the property of a coil that opposes any change in the amount of current flowing through it. If the current in the coil is increasing, the self-induced emf is set up in such a direction so as to oppose the rise of current. Similarly, if the current in the coil is decreasing, the self-induced emf will be in the same direction as the applied voltage.

Inductance is defined as the ratio of flux linkage to the current flowing through the coil. The unit of inductance is termed the *henry* and is represented by the symbol H. *A coil is said to have an inductance of 1 henry if a current of 1 ampere when flowing through it produces flux linkages of 1 weber-turn in it.*

The circuit symbol of inductor is shown in Figure 1.26



Figure 1.26: Inductor symbol

If I is the current, N is the number of turns and ϕ is the flux linking with the coil,then inductance L is:

$$L = \frac{N\phi}{I} \tag{1.13}$$

1.3.2 Analysis of a purely resistive circuit

Consider a circuit having resistance R ohms connected across the terminals of an a.c. supply, as in Figure.1.27a. In the circuit shown in Figure.1.27a, R is a pure resistance to which an alternating voltage v = V_m sinωt is applied, due to which an alternating current i flows through it.

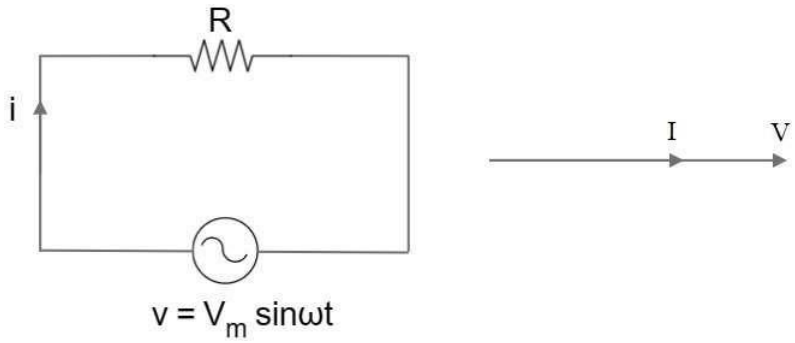


Figure 1.27: a) Pure resistive circuit b) Phasor Diagram

By Ohms law,

$$i = \frac{v}{R} = \frac{V_m sin\omega t}{R} = I_m sin\omega t \text{ where.} I_m = V_m/R \tag{1.14}$$

By observing the equations for voltage, v=$V_m$sinωt and i=I_m sinωt, it can be concluded that the current is in phase with the voltage. Vectorially the r.m.s values of voltage and current are represented as shown in Figure.1.27b.
The instantaneous power consumed by the resistance is given by.

$$p = v \times i = V_m sin\omega t I_m sin\omega t = V_m I_m sin^2\omega t$$

$$p = V_m I_m \frac{(1 - cos2\omega t)}{2} = \frac{1}{2} V_m I_m - \frac{1}{2} V_m I_m cos2\omega t \tag{1.15}$$

Equation 1.15 consists of two parts. The second part $\frac{1}{2} V_m I_m cos2\omega t$ is a periodically varying quantity whose frequency is two times the frequency of the applied voltage and its average value over a period of time is zero. Power is a scalar quantity and hence only its average value has to be taken into account. Hence the power consumed by the resistance is only due to the first part $\frac{1}{2} V_m I_m$.
$\therefore$ Average Power,

$$P = \frac{1}{2} V_m I_m = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} = V_{rms} I_{rms} \tag{1.16}$$

The waveforms of v, i and p are as shown in the Figure.1.28 below.

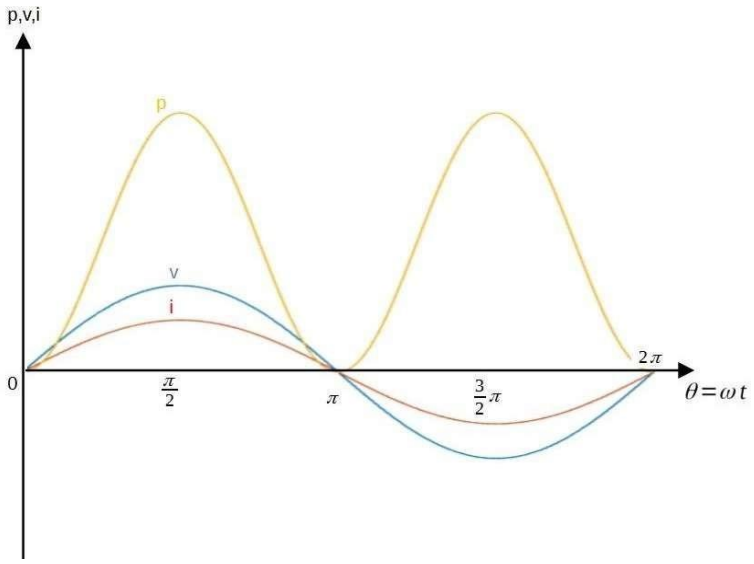


Figure 1.28: a) Waveforms of voltage, current and instantaneous power in pure resistive circuit

1.3.3 Analysis of a purely inductive circuit

Consider a circuit with a coil of pure inductance L henry connected across the terminals of an a.c. supply, as in Figure.1.29a. Due to the applied alternating voltage

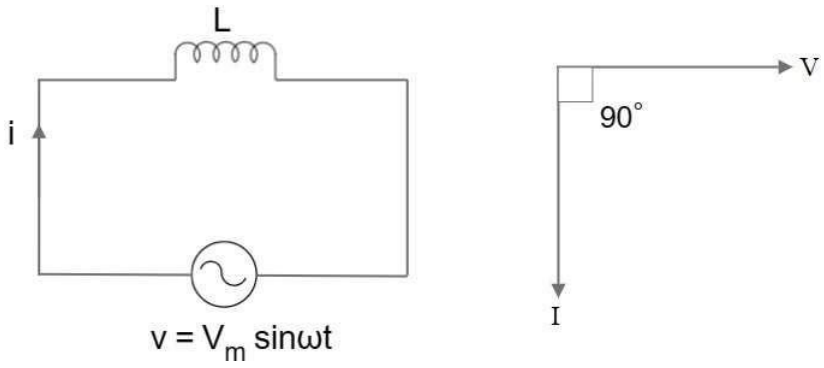


Figure 1.29: a) Pure inductive circuit b) Phasor Diagram

v=$V_m$sinωt, an alternating current i flows through it. This alternating current produces an alternating flux, which links the coil and hence an e.m.f e is induced in it, which opposes the applied voltage and is given by,

$$e = -L\frac{di}{dt} = -v$$

$$\therefore v = L\frac{di}{dt} \tag{1.17}$$

$$di = \frac{v}{L}dt = \frac{1}{L}V_m sin\omega t \quad dt \tag{1.18}$$

Integrating the equation 1.17, we get

$$\begin{aligned} i &= \frac{V_m}{L}\int sin\omega t \cdot dt \\ &= \frac{V_m}{\omega L}(-cos\omega t) \\ &= \frac{V_m}{\omega L}sin(\omega t - \frac{\pi}{2}) \end{aligned}$$

$$i = I_m sin(\omega t - \frac{\pi}{2}) \tag{1.19}$$

Where $I_m = \frac{V_m}{\omega L} = \frac{V_m}{X_L}$
$X_L = \omega L = 2\pi$fL = Inductive reactance in Ohms
By observing the equations for voltage and current, we find that the current lags the voltage by an angle $\pi/2$. Vectorial representation of the r.m.s values of voltage and

current is as shown in Figure 1.29b.
The instantaneous power is given by,

$$p = v \cdot i = V_m sin\omega t \cdot I_m sin(\omega t - \frac{\pi}{2})$$

$$p = V_m I_m sin\omega t(-cos\omega t) = -\frac{1}{2} V_m I_m sin2\omega t \tag{1.20}$$

The equation for p consists of a quantity which is periodically varying and having a frequency two times the frequency of the applied voltage and whose average value is zero.The power consumed by a pure inductor is zero, because the power is a scalar quantity and only its average value has to be considered. The power given to the pure inductance is stored in the form of an electromagnetic field. The waveforms of instantaneous voltage, current & power are as shown in Figure.1.30.

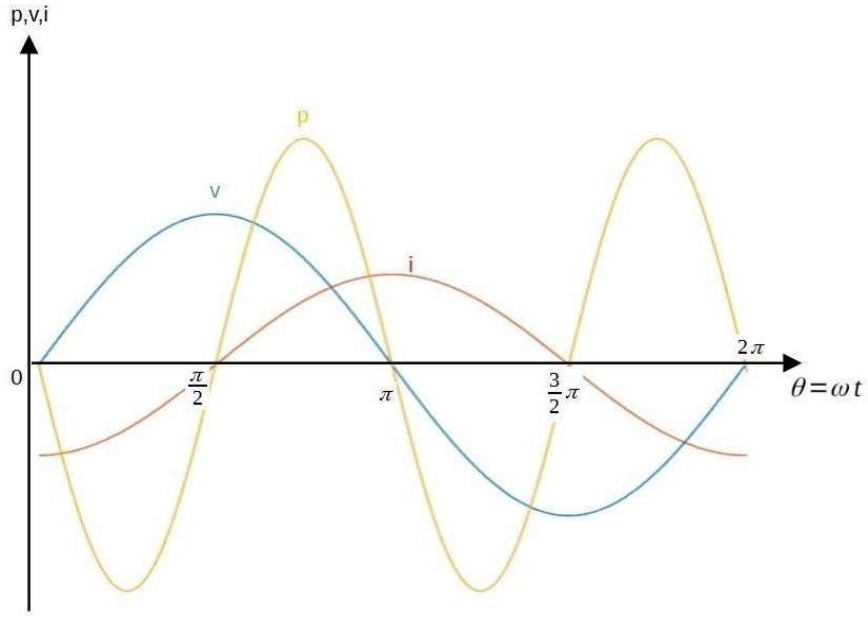


Figure 1.30: Waveforms of v,i and p for Pure inductive circuit

1.3.4 Analysis of a purely capacitive circuit:

Consider a capacitor of pure capacitance C, across which an alternating voltage v=$V_m sin\omega t$ is applied, due to which an alternating current i flows, charging the plates of the capacitor with a charge of q coulombs as shown in Figure 1.31a.

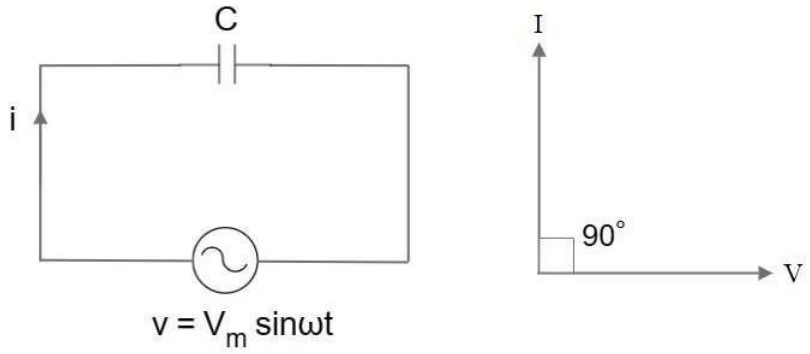


Figure 1.31: a) Pure capacitive circuit b) Phasor Diagram

$$
\begin{aligned}
i &= \frac{dq}{dt} = \frac{d(Cv)}{dt} = C\frac{dv}{dt} = C\frac{d}{dt}(V_m sin\omega t) \\
&= \omega C V_m cos\omega t = \frac{V_m}{1/\omega C} sin(\omega t + \pi/2) \\
&= \frac{V_m}{X_C} sin(\omega t + \pi/2)
\end{aligned}
$$

$$
i = I_m sin(\omega t + \pi/2) \tag{1.21}
$$

where, $I_m = \frac{V_m}{X_C}$ and $X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$ =Capacitive reactance in ohms.
By observing the equations for voltage and current, we find that the current leads the voltage by an angle $\pi/2$. Vectorial representation of the r.m.s values of voltage and current is as shown in Figure 1.31b.
The instantaneous power is given by,

$$
p = v \quad i = V_m sin\omega t \cdot I_m sin(\omega t + \frac{\pi}{2})
$$

$$
p = V_m I_m sin\omega t cos\omega t = \frac{1}{2} V_m I_m sin2\omega t \tag{1.22}
$$

The equation for p consists of a quantity which is periodically varying and having a frequency two times the frequency of the applied voltage and whose average value is zero.The power consumed by a pure capacitor is zero, because the power is a scalar

quantity and only its average value has to be considered. The power given to the pure capacitance is stored in the form of an electrostatic field. The waveforms of v, i and p are as shown in Figure.1.32.

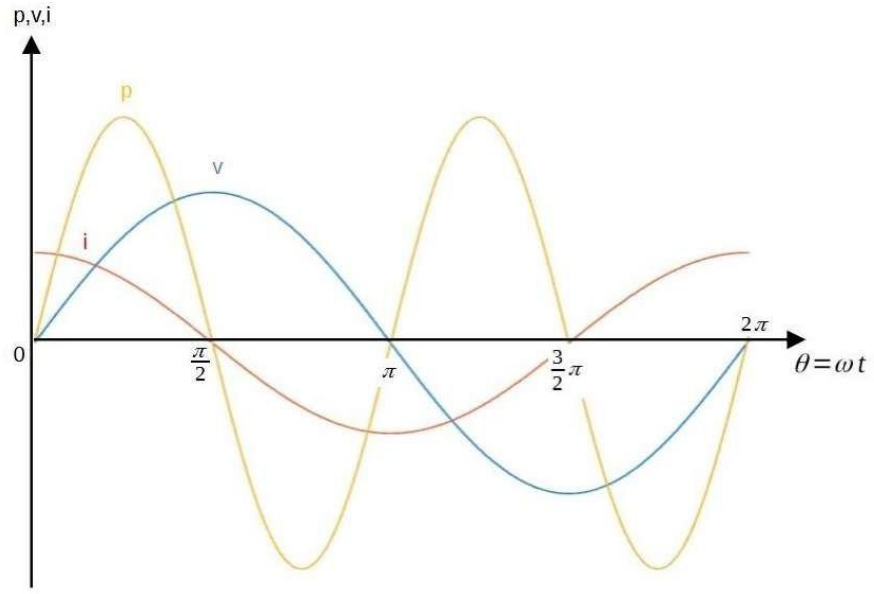


Figure 1.32: Waveforms of v,i and p for Pure capacitive circuit

1.3.5 Numericals

1. An AC Circuit consist of a pure resistance of 20 Ω and is connected across an AC supply of 230 V, 50Hz. Calculate current, power consumed, equation for voltage and current.
   Solution:
   I = V/R = 230/20 = 11.5A
   P = VI = 230 × 11.5 = 2645 W
   $V_m = \sqrt{2}$ V = 325.27 V
   $I_m = \sqrt{2}$ I = 16.26 A
   $\omega = 2\pi$f = 314Hz
   v = $V_m$sinωt = 325.27 sin314t V
   i = $I_m$sinωt = 16.26 sin314t A

2. A pure inductive coil allows a current of 20 A to flow from 230V, 50Hz supply. Calculate inductive reactance, inductance of the coil, power absorbed, equations for voltage and current.
   Solution:

Inductive reactance, X_L= V/I= 230/20 = 11.5 Ω
Coil inductance, L = X_L / 2πf = 11.5 / 2π*50 = 0.036H
Power absorbed = 0 (∵ Power absorbed in pure inductive circuit is zero)
Peak value of the voltage, V_m=$\sqrt{2}$V=$\sqrt{2}$*230=325.27V
Peak value of the current, I_m=$\sqrt{2}$ I = $\sqrt{2}$*20 =28.28 A
ω= 2πf = 314Hz
Equation of the voltage : v = $V_m$sinωt = 325.27 sin314t V
Equation of the current : i = $I_m$sin($\omega t - \frac{\pi}{2}$) = 28.28 sin(314 t-$\frac{\pi}{2}$) A

3. A 400μF capacitor is connected across a 230V, 50Hz supply. Calculate capacitive reactance, rms current, equations for the voltage and current.
Solution:
Capacitive reactance, X_C= 1 / (2πf C) =1 / (2π*50*400μ)= 7.95 Ω
RMS current, I = V/X_C= 230 / 7.95 = 28.93A
Peak value of the voltage, $V_m = \sqrt{2}V = \sqrt{2}$*230=325.27V
Peak value of the current, $I_m = \sqrt{2}I = \sqrt{2}$*28.93 =40.91 A
$\omega = 2\pi$f = 314Hz
Equation of the voltage: v = $V_m sin\omega t$= 325.27 sin314t V
Equation of the current: i = $I_m sin(\omega t + \pi/2)$= 40.91 sin(314t+π/2) A

1.3.6 Analysis of a R-L Series circuit

Consider an R-L series circuit as shown in Figure.1.33a below to which an alternating voltage of v = $V_m$sinωt is applied, due to which an alternating current i flows through the circuit. The vector diagram considering I as reference vector is as shown in Figure.1.33 b. The vector diagram consists of three voltages, V_R = I R which is in

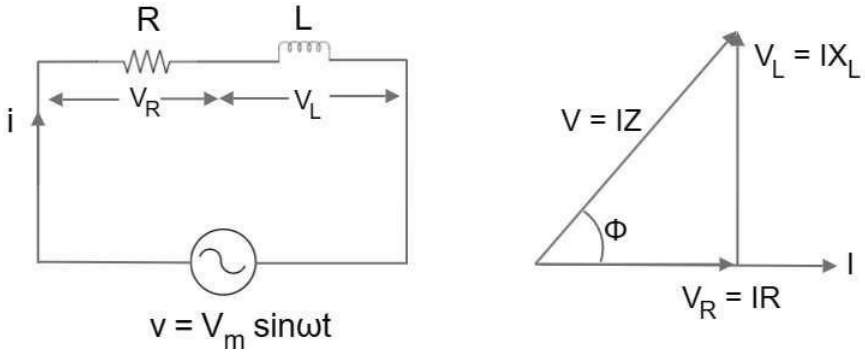


Figure 1.33: a) RL series circuit b) Phasor Diagram

phase with the current, V_L =I X_L which leads the current by 90°. The vector sum of these two voltages is applied voltage V=I Z. here, Z is the impedance of the circuit in ohms.

The impedance of an a.c circuit may be defined as the opposition offered for the flow of alternating current in the circuit. It is the phasor sum of resistance and reactance of the circuit. The impedance triangle for an R-L series circuit is shown in Figure 1.34.

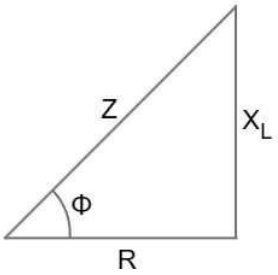


Figure 1.34: a) Impedance triangle

V = V_R + V_L (phasor sum)

$$V = \sqrt{V_R^2 + V_L^2} = \sqrt{I^2R^2 + I^2X_L^2}$$
$$= I\sqrt{R^2 + X_L^2} = IZ$$

where, Z= $\sqrt{R^2 + X_L^2}$ and $\phi = tan^{-1}(\frac{X_L}{R}) = cos^{-1}(\frac{R}{Z})$

From the phasor diagram shown in Figure.1.33 b. we observe that the current lags the voltage by an angle ϕ.
If v = V_m sinω t ,
Then, i= $I_m sin(\omega t - \phi)$
The instantaneous power is given by,

$$\begin{aligned} p = v \quad i &= V_m sin\omega t \cdot I_m sin(\omega t - \phi) \\ &= V_m I_m \frac{1}{2}[cos\phi - cos(2\omega t - \phi)] \end{aligned}$$

$$p = \frac{1}{2} V_m I_m cos\phi - \frac{1}{2} V_m I_m cos(2\omega t - \phi) \tag{1.23}$$

The second term in the equation is a periodically varying quantity, whose frequency is two times the frequency of the applied voltage and its average value is zero. As power is always an average value, only the first term represents the power consumed.

$$\begin{aligned} \therefore P = \frac{1}{2} V_m I_m cos\phi &= \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} cos\phi \\ &= VIcos\phi \end{aligned}$$

Where, cosϕ is known as the power factor of the circuit. The waveforms of v, i and p are shown in Figure.1.35. The areas of the +ve power lobes is more than the areas of the -ve power lobes, indicating that the power received by the circuit is more than the power returned by the circuit. Hence, the circuit consumes a net power given by,P =V I cos ϕ.

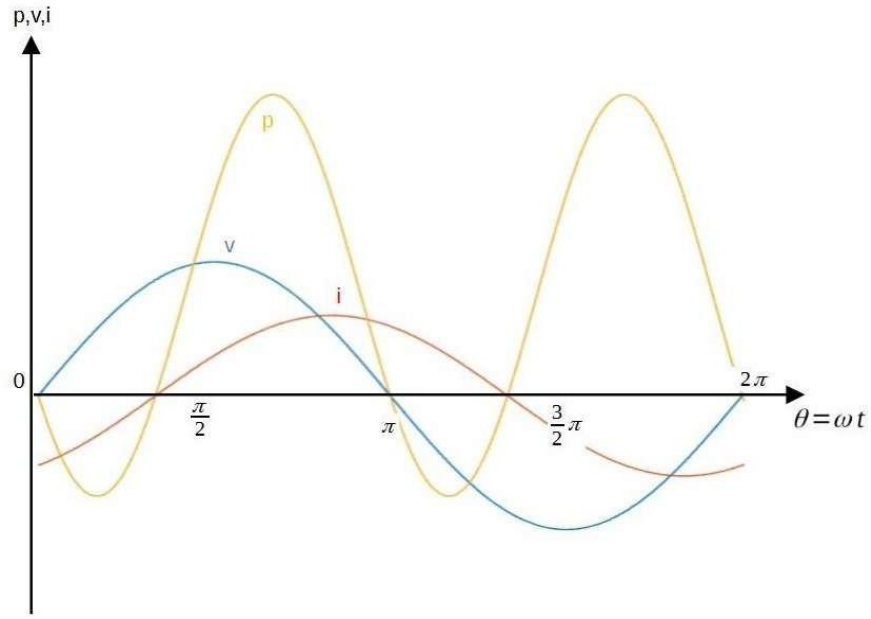


Figure 1.35: Waveforms of v,i and p for RL series circuit

1.3.7 Analysis of a R-C Series circuit

Consider an R-C series circuit as shown in Figure.1.36a. to which an alternating voltage of v = $V_m$sinωt is applied, due to which an alternating current i flows through the circuit. I is taken as the reference vector. The diagram consists of three voltages,

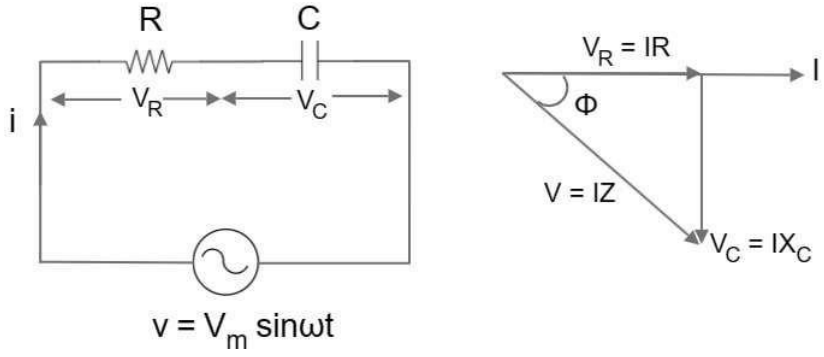


Figure 1.36: a) RC series circuit b) Phasor Diagram

V_R = I R which is in phase with the current, V_C =I X_C which lags the current by 90°. The vector sum of these two voltages is applied voltage V=IZ. Here, Z is the impedance of the circuit in ohms.

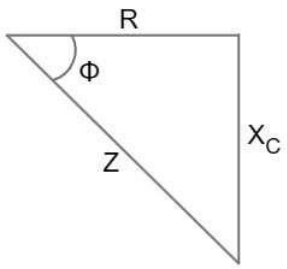


Figure 1.37: Impedance Diagram

V = $V_R + V_C$ (phasor sum)

$$V = \sqrt{V_R^2 + V_C^2} = \sqrt{I^2R^2 + I^2X_C^2}$$
$$= I\sqrt{R^2 + X_C^2} = IZ$$

where, Z= $\sqrt{R^2 + X_C^2}$ and $\phi = tan^{-1}(\frac{X_C}{R}) = cos^{-1}(\frac{R}{Z})$
The current leads the voltage by an angle ϕ.
If v = $V_m sin\omega t$, Then, i= $I_m sin(\omega t + \phi)$
The current in the circuit is given by,I=V/Z
The impedance triangle for an R-C series circuit is shown in Fig.1.37.
The instantaneous power is given by,

$$p = v \quad i = V_m sin\omega t \cdot Imsin(\omega t + \phi)$$
$$= V_m I_m \frac{1}{2}[cos(-\phi) - cos(2\omega t + \phi)]$$
$$p = \frac{1}{2}V_m I_m cos\phi - \frac{1}{2}V_m I_m cos(2\omega t + \phi) \tag{1.24}$$

The second term in the equation is a periodically varying quantity, whose frequency is two times the frequency of the applied voltage and its average value is zero. As power is always an average value, only the first term represents the power consumed.

$$\therefore P = \frac{1}{2}V_m I_m cos\phi = \frac{V_m}{\sqrt{2}}\frac{I_m}{\sqrt{2}}cos\phi$$
$$= VIcos\phi$$

The waveforms of v, i and p are shown in Figure.1.38.

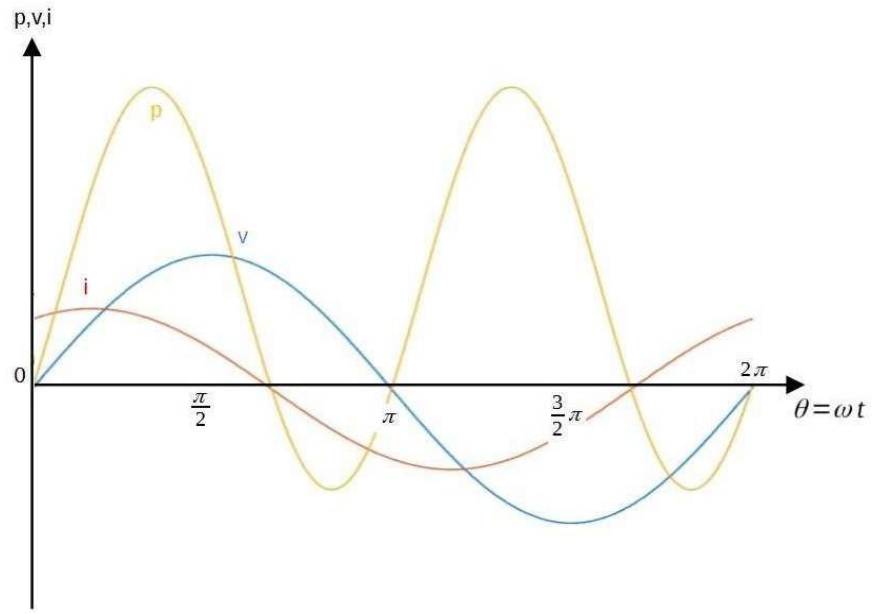


Figure 1.38: Waveforms of v,i and p for RC series circuit

1.3.8 Analysis of a R-L-C Series circuit

Consider an R-L-C series circuit as shown in Figure.1.39. to which an alternating voltage of v = $V_m$sinωt is applied, due to which an alternating current i flows through the circuit. Three cases of the circuit can be discussed here.

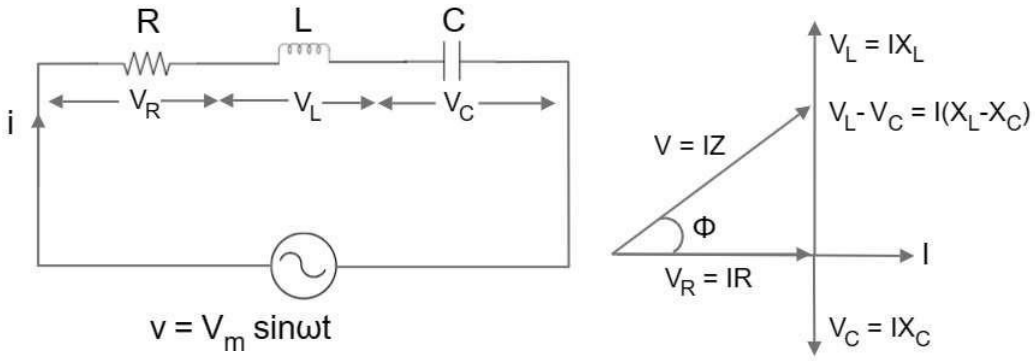


Figure 1.39: a) RLC series circuit b) Phasor Diagram for $X_L > X_C$

Case 1; When $X_L > X_C$

When the inductive reactance is more than the capacitive reactance. the vector diagram of the circuit as shown in Figure.1.39a. From the vector diagram, we observe that the current lags the voltage by an angle ϕ. The impedance triangle is shown in

Figure.1.39b

$$I = \frac{V}{Z}$$
$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

The circuit behaves similar to an R-L circuit.
If v = $V_m sin\omega t$,
Then, i= $I_m sin(\omega t - \phi)$
Hence it can be proved that the power consumed is given by P=VI cos ϕ.
Case 2; When $X_L < X_C$
When the inductive reactance is less than the capacitive reactance, the vector diagram of the circuit as shown in Figure.1.40a.From the vector diagram, we observe that the current leads the voltage by an angle ϕ

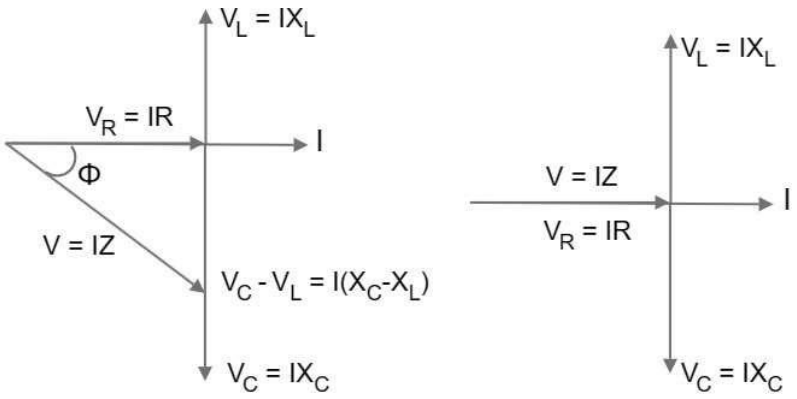


Figure 1.40: Phasor Diagram for a) $X_L < X_C$ b)$X_L = X_C$

$$I = \frac{V}{Z}$$
$$Z = \sqrt{R^2 + (X_C - X_L)^2}$$

The circuit is similar to an R-C circuit.
If v = $V_m sin\omega t$,
Then, i= $I_m sin(\omega t + \phi)$
Hence it can be proved that the power consumed is given by P=VI cos ϕ.
Case 3; When $X_L = X_C$
When the inductive reactance is equal to the capacitive reactance, the vector diagram of the circuit as shown in Figure. 1.40b. V_L and V_C cancel out each other. The current is in phase with the voltage and circuit behaves as a pure resistance circuit. Hence Z = R.

If v = $V_m sin\omega t$,
Then, i= $I_m sin\omega t$
Hence it can be proved that the power consumed is given by P=VI. which is proved in the case of a pure resistance circuit.

1.3.9 Numericals

1. A resistance of 10 Ω is connected in series with a pure inductance of 55mH and the circuit is connected to a 230V, 50Hz sinusoidal supply. Calculate the a) circuit current b) Phase angle
   Solution:
   Given data: R=10 Ω L= 55mH V=230V, f=50Hz
   a) Current I = V/Z
   $X_L = 2\pi fL = 2\pi \times 50 \times 55 \times 10^{-3} = 17.27\ \Omega$
   Impedance Z=$\sqrt{R^2 + X_L^2}$
   Substituting for X_L, Z= $\sqrt{10^2 + 17.27^2}$= 19.95 Ω
   ∴ Current I = 230/19.95=11.52A
   b) Phase angle $\phi = tan^{-1}\frac{X_L}{R} = tan^{-1}\frac{17.27}{10} = 60^\circ lag$ (∵ inductive circuit)

2. A capacitor of 16μ F takes a current of 2 A when the alternating voltage applied across it is 230 V. Calculate: (a) the frequency of the applied voltage; (b) the resistance to be connected in series with the capacitor to reduce the current in the circuit to 1 A at the same frequency; (c) the phase angle of the resulting circuit.
   Solution:
   (a)$X_C = \frac{V}{I} = \frac{230}{2} = 115\Omega$
   $X_C = \frac{1}{2\pi fC}$
   $\therefore f = \frac{1}{2\pi C X_C} = \frac{1}{2\pi \times 16 \times 10^{-6} \times 115} = 86.5$ Hz
   (b) When a resistance is connected in series with the capacitor,
   Z=$\frac{V}{I} = \frac{230}{1}$= 230 Ω
   Z=$\sqrt{R^2 + X_C^2} = \sqrt{R^2 + 115^2}$=230Ω
   hence R=199.18Ω
   (c) $\phi = cos^{-1}\frac{R}{Z} = cos^{-1}\frac{199.18}{230} = 30^\circ$ lead

3. A 230V, 50Hz AC supply is applied to a coil of 0.12 H inductance and 5 Ω resistance, the connection in series with a 12μF capacitance. Calculate the impedance, current, phase angle between voltage and current, power factor and

the power consumed.

Solution: $X_L = 2\pi fL = 2\pi \times 50 \times 0.12 = 37.69\ \Omega$

$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 50 \times 12 \times 10^{-6}} = 265.25\ \Omega$

$Z = \sqrt{R^2 + (X_C - X_L)^2} = 227.6\ \Omega$

$I = V/Z = 230/227.6 = 1.01A$

To find the phase angle of the resulting circuit:

$\tan\phi = (X_C - X_L)/R = 227.56/5 = 45.51$

Therefore $\phi = 88.74^\circ$ lead

$\cos\phi = R/Z = 5 / 227.6 = 0.0219$ lead

$P = VI \cos\phi = 230 \times 1.01 \times 0.0219 = 5.08$ W

4. A circuit having resistance of 6 Ω, an inductance of 0.075 Henry and a capacitance of 50μF in series is connected across a 100V, 50Hz supply. Calculate a) the impedance b) the current c) the voltage across R.L and C d) the phase difference between the current and the supply voltage.

   Solution:

   Data given: R=6Ω, L=0.075H, C=50μF, V=100V, f=50Hz

   $X_L = 2\pi fL = 23.56\ \Omega$

   $X_C = 1/2\pi fC = 63.66\ \Omega$

   $Z = \sqrt{R^2 + (X_C - X_L)^2} = 40.54\ \Omega$

   Current $I = V/Z = 100 / 40.54 = 2.466$ A

   Voltage across resistance $V_R = IR = 2.466 \times 6 = 14.76$ V

   Voltage across inductance $V_L = IX_L = 2.466 \times 23.56 = 58.09$ V

   Voltage across capacitance $V_C = IX_C = 2.466 \times 63.66 = 156.98$ V

   Phase difference $\phi = cos^{-1}\ (R/Z) = cos^{-1}\ (6/40.54) = 81.48^\circ$

1.3.10 Power

True Power and Reactive Power:

Power is consumed only in resistance and no power is consumed in pure inductor or pure capacitor. The power received from L and C in one quarter cycle is returned to the source in the next quarter cycle. This circulating power is called reactive power and does no useful work in the circuit. The power which is actually consumed in the circuit is called the true power or active power. Therefore current in-phase with voltage produces true power whereas the current 90° out of phase with voltage contributes to reactive power.

True Power= Voltage × current in phase with voltage
Reactive Power= Voltage × current 90° out of phase with voltage
The current can be resolved into two components:

1. I cosϕ -in phase with voltage
2. I sinϕ -90° out of phase with voltage

$\therefore$ True Power.P = V I Cosϕ Watts or kW
Reactive Power,Q= VI Sinϕ VAR(Volt Amp Reactive) or kVAR
Apparent power, S= VI VA(Volt-Amps) or kVA
From Figure. 1.41,

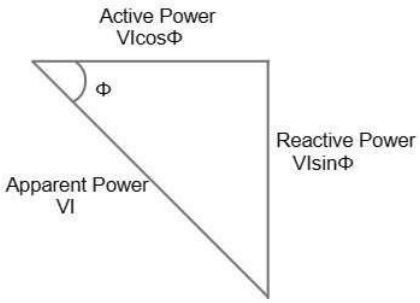


Figure 1.41: Power Triangle

$$S^2 = P^2 + Q^2$$
$$S = \sqrt{P^2 + Q^2}$$

1.3.11 Power Factor(p.f) of a circuit

The p.f of a circuit can be defined in the following three ways:-
The power factor of a circuit is the cosine of the angle between the voltage and the current. p.f = cos ϕ
The power factor of a circuit is the ratio of the resistance to the impedance of the circuit. p.f =R/Z.
The power factor of a circuit is the ratio of the real power to the apparent power. p.f = P/VI. The maximum value p.f is unity.